Chapter 4

Characterizing Analytic Functions

4.1 Power series

Theorem 4.1 (Power series expansion, Cauchy 1831)**.** Any analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open satisfies

$$\forall c \in \mathcal{D},\ \forall \mathcal{U}_R(c) \subseteq \mathcal{D},\ \forall z \in \mathcal{U}_R(c),\ f(z) = \sum_{n=0}^{\infty} a_n (z-c)^n, \tag{4.1}$$

where $\mathcal{U}_R(c)$ is the open disk with radius R and center c in Notation 1 and each a_n is uniquely determined as

$$a_n := \frac{f^{(n)}(c)}{n!} = \frac{1}{2\pi\mathbf{i}} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{(\zeta-c)^{n+1}} \mathrm{d}\zeta, \tag{4.2}$$

whose value does not depend on the choice of $\rho \in (0, R)$.

Proof. $|z-c| < |\zeta - c|$ holds for all $z \in \mathcal{U}_\rho(c)$. Then the geometric series yields

$$\tfrac{1}{\zeta-z} = \tfrac{1}{(\zeta-c)\left(1-\frac{z-c}{\zeta-c}\right)} = \textstyle\sum_{n=0}^{\infty} \tfrac{(z-c)^n}{(\zeta-c)^{n+1}}.$$

The Cauchy integral formula (Theorem 3.75) yields

$$\begin{aligned} f(z) &= \tfrac{1}{2\pi\mathbf{i}} \oint_{|\zeta-c|=\rho} \tfrac{f(\zeta)}{\zeta-z}\mathrm{d}\zeta \\ &= \tfrac{1}{2\pi\mathbf{i}} \oint_{|\zeta-c|=\rho} f(\zeta) \textstyle\sum_{n=0}^{\infty} \tfrac{(z-c)^n}{(\zeta-c)^{n+1}}\mathrm{d}\zeta \\ &= \textstyle\sum_{n=0}^{\infty} (z-c)^n \tfrac{1}{2\pi\mathbf{i}} \oint_{|\zeta-c|=\rho} \tfrac{f(\zeta)}{(\zeta-c)^{n+1}}\mathrm{d}\zeta \\ &= \textstyle\sum_{n=0}^{\infty} a_n (z-c)^n, \end{aligned}$$

where the third step follows from Theorems 1.77, 1.75 and D.178, and the last step from (4.2) and Theorem 3.85. Finally, the uniqueness of a_n follows from the uniqueness of the above derivation. □

Exercise 4.2 (Gutzmer's inequality)**.** With the assumptions of Theorem 4.1, show that for any $r \in (0, R)$,

$$\sum_{n=0}^{\infty} |a_n|^2 r^{2n} \le M(r)^2, \tag{4.3}$$

where $M(r)$ is defined in Corollary 3.86.

Exercise 4.3 (Alternative proof of Corollary 3.86)**.** Prove (3.50) by using (4.3). Furthermore, show that the equality in (3.50) holds for some $n_0 \in \mathbb{N}$ if and only if $f(z) = a_{n_0}(z-c)^{n_0}$.

Example 4.4. Theorem 4.1 applied to to $\mathrm{Log}(z)$ on $\mathbb{C}_-$ in (2.8) yields a power series expansion,

$$\mathrm{Log}(z) = \mathrm{Log}(c) + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{c^n n}(z-c)^n, \tag{4.4}$$

where we have used the fact $(\mathrm{Log} z)' = \frac{1}{z}$ (why?). By Theorem 1.79, the convergence radius is $|c|$. Interestingly, this example shows that the convergence radius may be bigger than the distance of the series center to the boundary of the domain of definition.

One needs to be careful about the convergence of a Taylor series. If $c \in \mathbb{C}_-$ satisfies $\mathrm{Re}\, c < 0$, then $\mathbb{C}_- \cap \mathcal{U}_r(c)$ might consist of two disjoint separable connected components, where the power series converge to different functions,

$$f(z) = \begin{cases} \mathrm{Log}(z) & \text{if } \mathrm{Im}\, c > 0; \\ \mathrm{Log}(z) + 2\pi\mathbf{i} & \text{if } \mathrm{Im}\, c < 0. \end{cases}$$

Example 4.5. $\sum_{n=1}^{\infty} \frac{z^n}{n^2}$ converges for $|z| = 1$, $\sum_{n=1}^{\infty} z^n$ diverges for $|z| = 1$, and the series

$$p(z) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n} \tag{4.5}$$

diverges for $z = -1$ and converges for all other z satisfying $|z| \le 1$. Why? The answers lie in Theorem 3.111 and the fact that $p(z) = \mathrm{Log}(1+z)$.

Example 4.6. Consider the geometric series

$$g(z) = \frac{1}{1-z^2} = \sum_{n=0}^{\infty} z^{2n}.$$

The divergence of $\sum_{n=0}^{\infty} z^{2n}$ for $|z| \ge 1, z \in \mathbb{R}$ is easily understood: $g(z)$ has two singularities at $z = \pm 1$. However the divergence of the geometric series

$$f(z) = \frac{1}{1+z^2} = \sum_{n=0}^{\infty} (-1)^n z^{2n}.$$

for $|z| \ge 1, z \in \mathbb{R}$ is perplexing. After all, $f(z)$ is always bounded for any given $z \in \mathbb{R}$. This puzzle is solved in the context of complex analysis by observing that $f(z)$ has two singularities at $z = \pm\mathbf{i}$. Indeed, $f(z) = g(\mathbf{i}z)$ and the two functions are really the same complex function with different coordinates.

Exercise 4.7 (Double series theorem of Weierstrass)**.** Let the power series

$$f_j(z) = \sum_{k=0}^{\infty} c_{jk}(z-c)^k \tag{4.6}$$

be convergent in the disk $\mathcal{U}_r(c)$ with $r > 0$. If the series $F := \sum_{j=0}^{\infty} f_j$ converges normally in $\mathcal{U}_r(c)$, show that F is analytic in $\mathcal{U}_r(c)$ and can be represented as

$$F(z) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^{\infty} c_{jk} \right) (z-c)^k. \tag{4.7}$$

Example 4.8. The following $\mathcal{C}^\infty$ function due to Cauchy

$$f(x) = \begin{cases} \exp(-\frac{1}{x^2}), & x \neq 0; \\ 0, & x = 0 \end{cases} \tag{4.8}$$

satisfies $f^{(n)}(0) = 0$ for all $n \in \mathbb{N}$, but it does not have a Taylor expansion at any neighborhood at 0.

Theorem 4.9 (Equivalence of analytic functions on open sets)**.** The following assertions are equivalent for a function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open.

(A) f is analytic at any point $z \in \mathcal{D}$.

(B) f is totally differentiable in the sense of real analysis ($\mathbb{C} = \mathbb{R}^2$) and satisfies the Cauchy–Riemann equations (2.34).

(C) f is continuous and for each triangle path $\langle z_1, z_2, z_3 \rangle$ whose convex hull is contained in $\mathcal{D}$ we have

$$\oint_{\langle z_1, z_2, z_3 \rangle} f(\zeta)\mathrm{d}\zeta = 0.$$

(D) f locally admits primitives, i.e., for each point $c \in \mathcal{D}$ there exists an open disk $\mathcal{U}(c) \subset \mathcal{D}$ such that $f|_{\mathcal{U}(c)}$ has a primitive.

(E) f is continuous, and for each disk $\mathcal{U}_\rho(c)$ satisfying $\overline{\mathcal{U}_\rho(c)} \subset \mathcal{D}$ we have

$$\forall z \in \mathcal{U}_\rho(c), \qquad f(z) = \frac{1}{2\pi\mathbf{i}} \oint_{|\zeta - c| = \rho} \frac{f(\zeta)}{\zeta - z} \mathrm{d}\zeta.$$

(F) f is locally representable as a convergent power series.

(G) f is representable as a power series in each open disk fully contained in $\mathcal{D}$.

Proof. We list the main points of the proof as follows.

- $(A) \Leftrightarrow (B)$: Cauchy–Riemann theorem 2.56;
- $(A) \implies (C)$: Cauchy theorem 3.55;
- $(C) \implies (A)$: Morera's theorem 3.93;
- $(C) \implies (D)$: Theorem 3.42 and Lemma 3.58;
- $(D) \implies (C)$: Theorem 3.42;
- $(A) \implies (E)$: Cauchy integral formula 3.75;
- $(E) \implies (F)$: Taylor theorem 4.1;
- $(F) \implies (A)$: Corollary 3.106;
- $(A) \implies (G)$: Taylor theorem 4.1;
- $(G) \implies (F)$: A universal statement implies a existential statement. □

Exercise 4.10. Let the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be convergent in $\mathcal{U}_r(0)$. Show that if f is nonconstant and $\sum_{n=2}^{\infty} n|a_n|r^{n-1} \leq |a_1|$, then f is injective.

Exercise 4.11. Let the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be convergent in $\mathcal{U}_R(0)$. Define $A(r) = \max_{|z|=r} \operatorname{Re} f(z)$, Show that for each $r \in (0, R)$ and $n \in \mathbb{N}^+$,

(a) $a_n r^n = \frac{1}{\pi} \int_0^{2\pi} [\operatorname{Re} f(re^{\mathbf{i}t})] e^{-\mathbf{i}nt} \mathrm{d}t$;

(b) $|a_n| r^n \leq 2A(r) - 2\operatorname{Re} f(0)$.

4.2 Elementary domains

Definition 4.12. A domain $\mathcal{D} \subset \mathbb{C}$ is called an *elementary domain* iff any analytic function defined on $\mathcal{D}$ has a global primitive in $\mathcal{D}$.

Lemma 4.13. Let $\mathcal{D}$ and Ω be two elementary domains. If $\mathcal{D} \cap \Omega$ is non-empty and connected, then $\mathcal{D} \cup \Omega$ is also an elementary domain.

Proof. Let $f : \mathcal{D} \cup \Omega \to \mathbb{C}$ be analytic. We need to show that f has a primitive. Since $\mathcal{D}$ and Ω are both elementary domains, f has a primitive $F_1 : \mathcal{D} \to \mathbb{C}$ on $\mathcal{D}$ and a primitive $F_2 : \Omega \to \mathbb{C}$ on Ω. The difference $F_1 - F_2$ must be locally constant in $\mathcal{D} \cap \Omega$, and therefore constant since $\mathcal{D} \cap \Omega$ is connected. By addition of a constant if necessary, we can assume WLOG that $F_1|_{\mathcal{D}\cap\Omega} = F_2|_{\mathcal{D}\cap\Omega}$. Hence the functions F_1 and F_2 now glue to a single function $F : \mathcal{D} \cup \Omega \to \mathbb{C}$. □

Corollary 4.14. Slit annuli are elementary domains.

Proof. By Example 3.63, the ring sector is a star-shaped domain in the sense of Definition 3.62 and thus an elementary domain. By Lemma 4.13, the union of two ring sectors are also an elementary domain if their intersection is non-empty and connected. Hence a ring sector can be extended to a slit annulus while retaining the property of elementary domains. □

Corollary 4.15. Let $\mathcal{D}_1 \subset \mathcal{D}_2 \subset \cdots$ be an increasing sequence of elementary domains. Then their union $\cup_{n=1}^{\infty} \mathcal{D}_n$ is also an elementary domain.

Proof. This follows directly from Lemma 4.13. □

Lemma 4.16. Suppose $\mathcal{D} \subset \mathbb{C}$ is an elementary domain and $\varphi : \mathcal{D} \to \mathcal{D}^*$ a globally conformal map in the sense of Definition 2.38. Then $\mathcal{D}^*$ is also an elementary domain.

Proof. It suffices to show that any analytic function $f_* : \mathcal{D}^* \to \mathbb{C}$ has a primitive F_* satisfying $F_*' = f_*$. The condition of φ being globally conformal, Definition 2.38, Theorem 2.39, Theorem 2.74, and Theorem 3.85 imply that φ, φ^{-1}, and φ' are all analytic. Hence $(f_* \circ \varphi)\varphi' : \mathcal{D} \to \mathbb{C}$ is analytic. Since $\mathcal{D}$ is an elementary domain, there exists

$F : \mathcal{D} \to \mathbb{C}$ such that $F' = (f_* \circ \varphi)\varphi'$. Hence we can construct $F_* : \mathcal{D}^* \to \mathbb{C}$ by $F_* = F \circ \varphi^{-1}$ such that

$$(F \circ \varphi^{-1})' = (F' \circ \varphi^{-1})\frac{1}{\varphi'} = f_*.$$ □

Exercise 4.17. Show that the sickle-shaped domain

$$\mathcal{D} = \mathcal{U}_1(0) \setminus \overline{\mathcal{U}_{\frac{1}{2}}\left(\frac{1}{2}\right)} = \left\{z \in \mathbb{C} : \ |z| < 1, \left|z - \frac{1}{2}\right| > \frac{1}{2}\right\}$$

is an elementary domain.
Hint: Consider the conformal map $z \mapsto \frac{1}{1-z}$. What is the image of $\mathcal{D}$ under this map?

4.3 Zeros of analytic functions

Definition 4.18. A point $a \in \Omega$ is called a *zero* or a *root* of an analytic function $f : \Omega \to \mathbb{C}$ iff $f(a) = 0$. A zero point a of f is of *multiplicity* m iff $\exists m \in \mathbb{N}^+$ such that

$$f^{(m)}(a) \neq 0 \text{ and } \forall j = 0, 1, \ldots, m-1, \ f^{(j)}(a) = 0.$$

In particular, a zero of multiplicity 1 is a *simple zero* or a *simple root*.

Example 4.19. The zeros of $f(z) = \sin z - 1$ form the set $\left\{\frac{\pi}{2} + 2k\pi : k \in \mathbb{Z}\right\}$. Each zero is of multiplicity 2 and thus a critical point of f.

Lemma 4.20. Let Ω be a domain and $a \in \Omega$ a zero point of an analytic function $f : \Omega \to \mathbb{C}$. Then the multiplicity of f at a is m if and only if $f(z) = (z-a)^m\varphi(z)$ where $\varphi(a) \neq 0$ and φ is analytic in $\mathcal{U}_r(a)$ for some $r > 0$.

Proof. This follows directly from Theorem 4.1. □

Exercise 4.21. For any analytic function $f : \mathcal{U}_R(0) \to \mathbb{C}$ with $R > 0$, show $f \equiv 0$ if there exist constants $\rho \in (0, 1)$ and $C > 0$ such that $|f(\frac{1}{n})| \leq C\rho^n$ holds for sufficiently large $n \in \mathbb{N}$.

4.3.1 Roots of a nonzero analytic function are discrete

Definition 4.22. A subset $\mathcal{M} \subseteq \mathcal{X}$ in a metric space $(\mathcal{X}, d)$ is *discrete* iff

$$\forall x \in \mathcal{X}, \exists \epsilon > 0 \text{ s.t. } \forall y \in \mathcal{M} \setminus \{x\}, \quad d(x, y) > \epsilon. \tag{4.9}$$

Such a set consists of *isolated points*.

Lemma 4.23. A subset $\mathcal{M} \subseteq \mathcal{X}$ is *discrete* in $\mathcal{X}$ if and only if there is no limit point of $\mathcal{M}$ in $\mathcal{X}$.

Proof. By Definition 2.16, the existence of a limit point x of $\mathcal{M}$ in $\mathcal{X}$ implies

$$\exists x \in \mathcal{X} \text{ s.t. } \forall \epsilon > 0, \exists y \in \mathcal{M} \text{ s.t. } d(x, y) \in (0, \epsilon).$$

The negation of the above statement yields (4.9). □

Example 4.24. The set $\mathcal{M} := \left\{\frac{1}{n} : n \in \mathbb{N}^+\right\}$ is not discrete in $\Omega = \mathbb{C}$, but is indeed discrete in $\Omega = \mathbb{C} \setminus \{0\}$.

Example 4.25. The zeros of $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} x^2 \sin\frac{1}{x} & \text{if } x \neq 0; \\ 0 & \text{if } x = 0, \end{cases} \tag{4.10}$$

are $0, \pm\frac{1}{n\pi}$ where $n \in \mathbb{N}^+$ and 0 is a limit point of the zeros.

Example 4.26. Combine (4.8) and (4.10) and we have a $\mathcal{C}^\infty$ function whose zeros have a limit point in $\mathbb{R}$,

$$f(x) = \begin{cases} \exp(-\frac{1}{x^2})\sin\frac{1}{x} & \text{if } x \neq 0; \\ 0 & \text{if } x = 0. \end{cases} \tag{4.11}$$

Theorem 4.27. The set of zeros of a nonzero analytic function $f : \Omega \to \mathbb{C}$ with $\Omega \subset \mathbb{C}$ open is discrete in Ω.

Proof. Let a be a zero of f with multiplicity m. Lemma 4.20 yields $f(z) = (z-a)^m\varphi(z)$ where $\varphi(a) \neq 0$ and φ is analytic in $\mathcal{U}_r(a)$ for some $r > 0$. The continuity of φ gives

$$\exists R > 0 \text{ s.t. } \forall z \in \mathcal{U}_R(a), \ \varphi(z) \neq 0.$$

Then the proof is completed by Definition 4.22. □

4.3.2 Local mapping behaviors

Theorem 4.28. Let $f : \mathcal{D} \to \mathbb{C}$ be an analytic function on an elementary domain, and $f(z) \neq 0$ for all $z \in \mathcal{D}$. There exists an analytic function $h : \mathcal{D} \to \mathbb{C}$ such that

$$f(z) = \exp(h(z)), \tag{4.12}$$

where the function h is called an *analytic branch of the logarithm* of f.

Proof. By Theorem 3.85, f being analytic implies that f' is also analytic. Hence $\frac{f'}{f}$ is also analytic and by Definition 4.12 it has a primitive, say F. Then the function $G(z) := \frac{\exp(F(z))}{f(z)}$ satisfies

$$\forall z \in \mathcal{D}, \ G'(z) = \frac{1}{f^2}\left[\frac{f'}{f}\exp(F)f - \exp(F)f'\right] = 0.$$

Hence for all $z \in \mathcal{D}$ we have $\exp(F(z)) = Cf(z)$ with $C \neq 0$. Set $C = \exp(c)$ and the following function satisfies (4.12):

$$h(z) = F(z) - c.$$ □

Corollary 4.29. Under the assumption of Theorem 4.28, there exists for any $n \in \mathbb{N}$ an analytic function $H : \mathcal{D} \to \mathbb{C}$ such that $H^n = f$.

Proof. Set $H(z) = \exp(\frac{1}{n}h(z))$ and use Theorem 4.28. □

Theorem 4.30 (Local behavior of an analytic function at a root)**.** For a nonconstant analytic function $f : \mathcal{D} \to \mathbb{C}$ where $\mathcal{D}$ is a domain, $0 \in \mathcal{D}$, and $f(0) = 0$, there exists $r > 0$ and a globally conformal map $g : \mathcal{U}_r(0) \to \mathbb{C}$ such that

$$\forall z \in \mathcal{U}_r(0), \ f(z) = [g(z)]^m \tag{4.13}$$

where m is the multiplicity of the root of f at 0.

Proof. Set $a = 0$ in Lemma 4.20 and we have

$$\exists R > 0 \text{ s.t. } \forall z \in \mathcal{U}_R(0), \quad f(z) = z^m \varphi(z), \tag{4.14}$$

where φ is analytic in $\mathcal{U}_R(0)$ and $\varphi(0) \neq 0$. The continuity of φ implies the existence of $\bar{r} \in (0, R]$ such that $\varphi(z) \neq 0$ for all $z \in \mathcal{U}_{\bar{r}}(0)$. By Corollary 4.29, there exists an analytic function h such that $h^m = \varphi$. Then (4.13) follows from

$$g(z) := zh(z).$$

Clearly, $g : \mathcal{U}_{\bar{r}}(0) \to \mathbb{C}$ is analytic. For g to be globally conformal in the sense of Definition 2.38, Theorem 2.39 implies that it suffices to show $g'(0) \neq 0$, because then Theorem 4.27 and the inverse function theorem would imply the existence of $\tilde{r} > 0$ such that g is a homeomorphism in $\mathcal{U}_{\tilde{r}}(0)$ and then we can set $r = \min(\bar{r}, \tilde{r})$. Theorem 4.1 gives $\varphi(z) = \sum_{j=0}^{\infty} a_{m+j} z^j$ and thus

$$f(z) = z^m \sum\nolimits_{j=0}^{\infty} a_{m+j} z^j, \tag{4.15}$$

where $a_m \neq 0$ follows from Definition 4.18. On the one hand, (4.15) yields $f^{(m)}(0) = m! a_m$. On the other hand, (4.13) gives $f^{(m)}(0) = m!(g'(0))^m$. Then $0 \neq a_m = (g'(0))^m$ implies $g'(0) \neq 0$. □

Exercise 4.31. Let l_1, l_2 be two rays that both start from the origin and have an angle $\theta \in (0, \frac{2\pi}{m})$. Find the angle between the image curves $f(l_1)$ and $f(l_2)$ at 0 by assuming the conditions in Theorem 4.30.

Corollary 4.32. A $\mathcal{C}^1$ function $f : \mathcal{D} \to \Omega$ with $\mathcal{D}, \Omega \subseteq \mathbb{R}^2$ open is globally conformal if and only if f is bijective and analytic when $\mathcal{D}, \Omega$ are interpreted as open subsets of $\mathbb{C}$.

Proof. The necessity follows directly from Definition 2.38 and Theorem 2.39. For the sufficiency, Theorem 4.30 implies that the algebraic multiplicity m in (4.13) be 1; otherwise f would not be injective, letting alone being bijective. Since g is globally conformal, $g'(0) \neq 0$, which gives $f'(0) \neq 0$. □

Corollary 4.33. If $\varphi : \mathcal{D} \to \Omega$ with $\mathcal{D}, \Omega \subseteq \mathbb{C}$ open is *biholomorphic*, i.e., bijective and analytic, then so is its inverse function $\varphi^{-1} : \Omega \to \mathcal{D}$.

Proof. The follows from Corollaries 2.75 and 4.32. □

Exercise 4.34. For a nonconstant analytic function $f : \mathcal{D} \to \mathbb{C}$ where $\mathcal{D}$ is a domain, $0 \in \mathcal{D}$, $f(0) = f'(0) = 0$, and $f''(0) \neq 0$, construct two curves Γ_1 and Γ_2 so that

(a) both Γ_1 and Γ_2 pass through 0,

(b) Γ_1 and Γ_2 are orthogonal at 0,

(c) $f(\Gamma_1)$ is a subset of $\mathbb{R}$ and has a minimum at 0,

(d) $f(\Gamma_2)$ is a subset of $\mathbb{R}$ and has a maximum at 0.

4.3.3 The identity theorem

Theorem 4.35 (Identity)**.** For two analytic functions $f, g : \mathcal{D} \to \mathbb{C}$ defined on a nonempty domain $\mathcal{D}$, the following statements are equivalent:

(a) $f = g$.

(b) The coincidence set $\{z \in \mathcal{D} : f(z) = g(z)\}$ has a limit point in $\mathcal{D}$.

(c) $\exists a \in \mathcal{D}$ s.t. $\forall n \in \mathbb{N}$, $f^{(n)}(a) = g^{(n)}(a)$.

Proof. The equivalence of (a) and (b) follows from applying Theorem 4.27 to $f - g$ and that of (a) and (c) from applying Theorems 4.27 and 4.1. □

Lemma 4.36. If the product of two analytic functions on a domain $\mathcal{D}$ vanishes identically, then one of the two functions vanishes identically on $\mathcal{D}$.

Proof. Suppose $f(z)g(z) = 0$ for all $z \in \mathcal{D}$. We show $f \neq 0$ implies $g = 0$. $f \neq 0$ gives $\exists a \in \mathcal{D}$ s.t. $f(a) \neq 0$. Then the continuity of f yields

$$\exists r > 0 \text{ s.t. } \forall z \in \mathcal{U}_r(a), \ f(z) \neq 0.$$

Hence $fg = 0$ and Theorem 4.27 imply $g(z) = 0$ for all $z \in \mathcal{U}_r(a)$. The proof is completed by Theorem 4.35. □

Exercise 4.37. Determine whether there exist analytic functions $f, g : \mathcal{U}_R(0) \to \mathbb{C}$ satisfying

(a) $f(\frac{1}{n}) = f(-\frac{1}{n}) = \frac{1}{n^2}$, $n \geq 1$;

(b) $g(\frac{1}{2n-1}) = g(\frac{1}{2n}) = \frac{1}{2^n}$, $n \geq 1$.

Exercise 4.38. Suppose an analytic function f on a domain $\mathcal{D}$ satisfies

$$\forall z \in \mathcal{D}, \ \exists n \in \mathbb{N} \text{ s.t. } f^{(n)}(z) = 0.$$

Show that f is a polynomial.
Hint: Write $\mathcal{D} = \cup_{n=0}^{\infty} g_n^{-1}(\{0\})$, where $g_n(z) = f^{(n)}(z)$.

Notation 5. $H(\mathcal{D})$ denotes the set of analytic functions on a nonempty open subset $\mathcal{D} \subset \mathbb{C}$. $\mathcal{O}(\mathcal{D})$ denotes the set of analytic functions on a domain $\mathcal{D} \subset \mathbb{C}$.

Lemma 4.39. $H(\mathcal{D})$ is a commutative ring with unity.

Proof. By Definition G.32, $H(\mathcal{D})$ is clearly a ring with unity since the constant function $\mathbf{1}$ is the multiplication identity. The commutativity of the functions follows from that of complex numbers. □

Exercise 4.40. If $\mathcal{D}$ is not connected, then $H(\mathcal{D})$ might have divisors of zero. Give one such example.

Theorem 4.41. $H(\mathcal{D})$ is an integral domain if and only if $\mathcal{D}$ is a domain.

Proof. For a nonempty open set $\mathcal{D} \subseteq \mathbb{C}$, Lemma 4.39 states that $H(\mathcal{D})$ is a commutative ring with unity, where the unity $\mathbf{1}$ is different from the additive identity $\mathbf{0}$.

The sufficiency follows from Definition G.40 and the fact of $H(\mathcal{D})$ having no divisors of zero, thanks to Lemma 4.36 and Definition G.34.

As for the necessity, the only difference between an integral domain and a commutative ring with unity $\mathbf{1} \neq \mathbf{0}$ is that the integral domain contains no divisors of zero. Suppose $\mathcal{D}$ in Notation 5 is not connected. Then Exercise 4.40 leads to a contradiction, which completes the proof. □

Example 4.42. The unit disk $\mathbb{E}$ is a domain, but the analytic function $f(z) = z$ is zero at $z = 0$ and thus does not have a multiplicative inverse in $\mathcal{O}(\mathbb{E})$.

4.3.4 The open mapping theorem

Definition 4.43. A function is *open* iff it maps open sets to open sets.

Example 4.44. The function $f(x) = \sin x$ is $\mathcal{C}^\infty$ and $\mathbb{R}$ is open, but $\sin(\mathbb{R}) = [-1, 1]$ is not open.

Theorem 4.45 (Open mapping)**.** Any nonconstant analytic function on a domain is open.

Proof. The function $p(z) = z^n$ maps an open disk of radius r onto to an open disk of radius r^n: $re^{\mathrm{i}\varphi} \mapsto r^n e^{\mathrm{i}n\varphi}$. At each point z of an open set Ω, there exists an open disk $\mathcal{U}_r(z)$ at z such that $\mathcal{U}_r(z) \subset \Omega$. Then Ω can be considered as the union of open disks and Definition 4.43 implies that p is open. The rest of the proof follows from Theorem 4.30 and the fact that the function g is a globally conformal map and thus a homeomorphism. □

Corollary 4.46. Let $\mathcal{D}$ be a domain and $f : \mathcal{D} \to \mathbb{C}$ be an analytic function. If $\operatorname{Re} f$ or $\operatorname{Im} f$ or $|f|$ is a constant function, then f is a constant function.

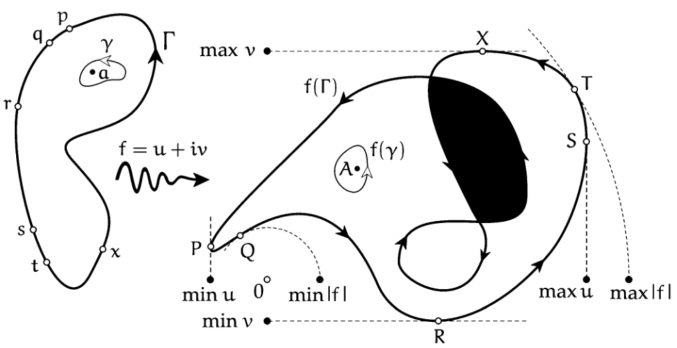
Proof. If $\operatorname{Re} f$ or $\operatorname{Im} f$ or $|f|$ is a constant, $f(\mathcal{D})$ cannot be an open set in $\mathbb{C}$, and thus Definition 4.43 and $\mathcal{D}$ being open imply that f is not an open map. The rest of the proof follows directly from the contraposition of Theorem 4.45. □

Exercise 4.47. Show that if $f : \mathcal{D} \to \mathcal{D}$ is analytic on a domain $\mathcal{D}$ and satisfies $f \circ f = f$, then f is either constant or the identity map.

4.3.5 Principles of maximum and minimum moduli

Theorem 4.48 (Maximum modulus)**.** Suppose the modulus of an analytic function $f : \mathcal{D} \to \mathbb{C}$ on a domain $\mathcal{D}$ reaches its maximum on $\mathcal{D}$. Then f is a constant.

Proof. Suppose $|f|$ reaches its maximum, say, at $a \in \mathcal{D}$. Then $f(U)$ is not open for any open ball $\mathcal{U}$ centered at a. Hence f is not open and the proof is then completed by the contraposition of Theorem 4.45. □



Corollary 4.49. Let $\mathcal{D}$ be a domain and $f : \mathcal{D} \to \mathbb{C}$ be an analytic function. For a compact set $K \subset \mathcal{D}$, the restriction $f|_K$ has a maximal modulus on K. Furthermore, the maximal modulus value occurs on the boundary of K.

Proof. This follows directly from Theorem 4.48. □

Exercise 4.50. Let $w_1, \ldots, w_n$ be points on the unit circle. Prove that there exists a point z on the unit circle such that the product of the distances from z to the points w_j is at least 1, i.e., $\prod_{j=1}^n |z - w_j| \geq 1$. Conclude that there exists a point w on the unit circle such that $\prod_{j=1}^n |w - w_j| = 1$.

Exercise 4.51. Show that if the real part of an entire function f is bounded, then f is constant.

Theorem 4.52 (Minimum modulus)**.** For a nonconstant analytic function $f : \mathcal{D} \to \mathbb{C}$ on a domain $\mathcal{D}$, a (local) minimum modulus of f at $a \in \mathcal{D}$ implies that a is a root of f, i.e., $f(a) = 0$.

Proof. Suppose $f(a) \neq 0$. Then $|f(a)| > 0$. Since $|f|$ reaches its local minimum at a, $f(U)$ is not open for any open ball $\mathcal{U}$ centered at a. Hence $f(\mathcal{D})$ is not open. But the contraposition of Theorem 4.45 implies that f is constant, which contradicts the given condition. □

Exercise 4.53. Prove the fundamental theorem of algebra by Theorem 4.52.

Exercise 4.54. Let f be nonconstant and analytic in an open set containing $\overline{\mathbb{E}}$. Show that if $|f(z)| = 1$ for all $|z| = 1$, then $\mathbb{E} \subseteq f(\mathbb{E})$.
Hint: One must show that $f(z) = w_0$ has a root for every $w_0 \in \mathbb{E}$. Then it suffices to consider the case of $w_0 = 0$. Why?

Exercise 4.55. Show that there is no analytic function f in $\mathbb{E}$ that extends continuously to $\partial\mathbb{E}$ and satisfies $f(z) = \frac{1}{z}$ for all $z \in \partial\mathbb{E}$.

4.3.6 The Schwarz lemma

Lemma 4.56 (Schwarz 1869)**.** An analytic function $f : \mathbb{E} \to \overline{\mathbb{E}}$ with $f(0) = 0$ must satisfy $|f'(0)| \leq 1$ and

$$\forall z \in \mathbb{E}, \quad |f(z)| \leq |z|. \tag{4.16}$$

Furthermore, if any of the two equalities holds, i.e., if $|f'(0)| = 1$ or $|f(a)| = |a|$ for some $a \neq 0$, then f must be a rotation, i.e., $f(z) = \zeta z$ for some $\zeta \in \mathbb{C}$ with $|\zeta| = 1$.

Proof. By Theorem 4.1, we can write $f = \sum_{j=0}^\infty a_j z^j$. Then $f(0) = 0$ implies $a_0 = 0$. The function $g : \mathbb{E} \to \mathbb{C}$ given by

$$g(z) := \sum\nolimits_{j=1}^\infty a_j z^{j-1}$$

satisfies $f(z) = zg(z)$ and $f'(0) = g(0)$ and is analytic. Then the condition $f(z) \in \overline{\mathbb{E}}$ implies $|f(z)| \leq 1$, which gives $|g(z)| \leq \frac{1}{|z|}$. Furthermore, Corollary 4.49 gives

$$\forall r \in (0, 1),\ \forall z \in \overline{\mathcal{U}_r(0)},\ |g(z)| \leq \frac{1}{r}.$$

Pass to the limit $r \to 1$ and we have $|g(z)| \leq 1$, which implies $|f'(0)| \leq 1$ and (4.16).

Suppose $f'(0) = 1$ or $|f(a)| = |a|$ for some $z = a \neq 0$. Then $|g(0)| = 1$ or $|g(a)| = 1$ and thus $g(z)$ attains its maximum modulus on $\mathbb{E}$. By Theorem 4.48, $g(z)$ is a constant on $\mathbb{E}$: $g(z) \equiv \zeta$, which completes the proof. □

Exercise 4.57 (Alternative proof of Liouville's theorem). Prove Theorem 3.95 by Lemma 4.56.

Exercise 4.58. For biholomorphic functions $f, g : \mathbb{E} \to \mathbb{E}$ with $f(0) = g(0)$ and $f'(0) = g'(0)$, show that $f = g$.

Exercise 4.59 (Study 1913). For a biholomorphic function $f : \mathbb{E} \to \Omega$, show that if Ω is convex, then for any $r \in (0, 1)$, the image $\Omega_r = f(\mathcal{U}_r(0))$ is also convex.
Hint: For two points $|a| \le |b| < r$ and $t \in [0, 1]$, consider the map $g_t(z) = (1 - t)f(\frac{az}{b}) + tf(z)$.

4.3.7 The automorphism group of $\mathbb{E}$

Definition 4.60. The *group of automorphisms* of a domain $\mathcal{D} \subseteq \mathbb{C}$, denoted by $\mathrm{Aut}(\mathcal{D})$, is the group of all globally conformal maps (or equivalently all biholomorphic functions) $\mathcal{D} \to \mathcal{D}$ with function composition.

Corollary 4.61. A biholomorphic function $\varphi_0 : \mathbb{E} \to \mathbb{E}$ with $\varphi_0(0) = 0$ must be a pure rotation, i.e.,

$$\exists \zeta \in \mathbb{C} \text{ with } |\zeta| = 1 \text{ s.t. } \varphi_0(z) = \zeta z. \tag{4.17}$$

Proof. By Corollary 4.33, the inverse φ_0^{-1} exists and is also analytic. Applying Lemma 4.56 to both φ_0 and φ_0^{-1} gives

$$|\varphi_0(z)| \le |z| = \left|\varphi_0^{-1}(\varphi_0(z))\right| \le |\varphi_0(z)|.$$

The rest follows from the last sentence of Lemma 4.56. ☐

Lemma 4.62. For any fixed $a \in \mathbb{E}$, the map $\varphi_a : \mathbb{E} \to \mathbb{E}$ given by

$$\varphi_a(z) := \frac{z - a}{\overline{a}z - 1} \tag{4.18}$$

is a bijective map of the unit disk onto itself which satisfies

(a) $\varphi_a(a) = 0$,

(b) $\varphi_a(0) = a$,

(c) $\varphi_a^{-1} = \varphi_a$.

In particular, φ_a is in both directions analytic.

Proof. First, φ_a in (4.18) is well defined as the denominator has no zeros in $\mathbb{E}$. Second, φ_a is injective because

$$z_1 \ne z_2 \implies \varphi_a(z_1) - \varphi_a(z_2) = \frac{(z_2 - z_1)(1 - |a|^2)}{(\overline{a}z_1 - 1)(\overline{a}z_2 - 1)} \ne 0.$$

Third, φ_a is surjective because

$$|\varphi_a(z)| < 1 \Leftrightarrow |z-a|^2 < |\overline{a}z-1|^2 \Leftrightarrow (1-|a|^2)(1-|z|^2) > 0.$$

Finally, (a), (b), and (c) follow directly from (4.18). ☐

Exercise 4.63. Consider an analytic function $f : \mathbb{E} \to \mathbb{E}$ that is not the identity function.

(a) Show that f has at most one fixed point.

(b) Give an example of f having no fixed points.

Exercise 4.64. For an analytic function $f : \mathbb{E} \to \mathbb{E}$, show that if there exists a point $a \in \mathbb{E} \setminus \{0\}$ such that $f(0) = a$ and $f(a) = 0$, then $f = \varphi_a$, where φ_a is as defined in (4.18).

Theorem 4.65. Any $f \in \mathrm{Aut}(\mathbb{E})$ is of the form

$$f(z) = e^{\mathbf{i}\theta}\frac{z - a}{\overline{a}z - 1}, \tag{4.19}$$

where $\theta \in [0, 2\pi)$ and $a \in \mathbb{E}$ depend only on f.

Proof. Since f is bijective, it maps one and only one point $a \in \mathbb{E}$ to 0, i.e., $a = f^{-1}(0)$. By Lemma 4.62(b), the map $g := f \circ \varphi_a$ is biholomorphic and satisfies $g(0) = 0$. The rest follows from Corollary 4.61 and Lemma 4.62(c). ☐

Theorem 4.66 (Schwarz–Pick). Any nonconstant analytic function $f : \mathbb{E} \to \mathbb{E}$ satisfies

$$\forall z \in \mathbb{E}, \qquad \frac{|f'(z)|}{1 - |f(z)|^2} \le \frac{1}{1 - |z|^2}, \tag{4.20}$$

where the equality holds for some $z \in \mathbb{E}$ if and only if $f \in \mathrm{Aut}(\mathbb{E})$.

Exercise 4.67. Prove Theorem 4.66.

4.3.8 Brouwer's fixed point theorem

Definition 4.68. A *fixed point* of a map $f : X \to Y$ is a point $p \in X$ that satisfies $f(p) = p$.

Definition 4.69. A subspace X of a topological space Y is a *retract* of Y if there is a continuous function $r : Y \to X$ with $r(x) = x$ for all $x \in X$; such a function is called a *retraction*.

Definition 4.70. The *inclusion* $i_A : A \hookrightarrow X$ of a topological space X and a subset A of X is

$$\forall x \in A, \qquad i_A(x) = x. \tag{4.21}$$

Lemma 4.71. A continuous map $r : Y \to X$ is a retraction if and only if $r \circ i = 1_X$ where $i : X \hookrightarrow Y$ is an inclusion.

Proof. For the sufficiency, $r \circ i = 1_X$ implies

$$\forall x \in X, \qquad r(x) = r \circ i(x) = 1_X(x) = x.$$

Hence, r is a retraction. As for the necessity, $r \circ i$ maps X to X and $\forall x \in X$, $r \circ i(x) = r(x) = x$, hence $r \circ i = 1_X$. ☐

Lemma 4.72. For $n \ge 0$, the n-sphere $\mathbb{S}^n$ cannot be a retract of the $(n + 1)$-ball $\overline{\mathbb{D}^{n+1}}$.

Proof. See [Rotman, 1988, p. 3]. ☐

Theorem 4.73 (Brouwer's fixed point). A continuous function $f : \overline{\mathbb{D}^n} \to \overline{\mathbb{D}^n}$ must have a fixed point $\mathbf{x}^* = f(\mathbf{x}^*)$.

Proof. Suppose that there exists $f : \overline{\mathbb{D}^n} \to \overline{\mathbb{D}^n}$ such that $f(\mathbf{x}) \ne \mathbf{x}$ for all $\mathbf{x} \in \overline{\mathbb{D}^n}$. Then we can determine a ray from $f(\mathbf{x})$ to $\mathbf{x}$, which intersects $\mathbb{S}^{n-1}$ at $\mathbf{y}$. Define $g : \overline{\mathbb{D}^n} \to \mathbb{S}^{n-1}$ by $g(\mathbf{x}) = \mathbf{y}$ and this construction implies that g is a retraction, but this contradicts Lemma 4.72. ☐

Exercise 4.74. The ray in the proof of Theorem 4.73 starts from $f(\mathbf{x})$. Can we start the ray from $\mathbf{x}$ and still get g as a retraction? Why?

Example 4.75. Let $f : \mathbb{D}^n \to \mathbb{R}^n$ be a homeomorphism between $\mathbb{D}^n$ and $\mathbb{R}^n$. For any $\mathbf{v} \neq \mathbf{0}$, the translation $T : \mathbb{R}^n \to \mathbb{R}^n$ given by $T(\mathbf{y}) = \mathbf{y} + \mathbf{v}$ has no fixed points, neither does the map $f^{-1} \circ T \circ f : \mathbb{D}^n \to \mathbb{D}^n$. For $n = 2$, a homeomorphism between $\mathbb{D}$ and $\mathbb{C}$ is $f(z) = \frac{z}{1-|z|}$ since its inverse $w \mapsto= \frac{w}{1+|w|}$ is also continuous.

4.4 Analytic continuation

Definition 4.76. Let $\mathcal{G}$ and $\mathcal{D}$ be two subsets of $\mathbb{C}$. An analytic function $F : \mathcal{G} \to \mathbb{C}$ is said to be an *analytic continuation* of another analytic function $f : \mathcal{D} \to \mathbb{C}$ iff $\mathcal{G} \supset \mathcal{D}$ and $F(z) = f(z)$ for all $z \in \mathcal{D}$.

Example 4.77. Corollary 3.66 is an example of analytic continuation for f from $\mathcal{D} \setminus \{z_*\}$ to $\mathcal{D}$.

Lemma 4.78. Let $\mathcal{D} \subseteq \mathbb{C}$ be a domain and $L \subset \mathbb{C}$ be a straight line. If a continuous function $f : \mathcal{D} \to \mathbb{C}$ is analytic at all points $z \in \mathcal{D} \setminus L$, then f is analytic on $\mathcal{D}$.

Proof. We use Morera's theorem 3.93 to prove this. WLOG, we can assume that L is the real axis $\mathbb{R}$; otherwise we consider the function $f \circ g$ where $g(z) = e^{\mathbf{i}\theta} z + c$ is a Euclidean isometry for any fixed θ and c given *a priori*. There are four possibilities with respect to the intersection of $\mathbb{R}$ to an arbitrary triangle $\Delta(z_1, z_2, z_3)$ in Definition 3.50.

(a) No intersections: (3.51) follows from the Cauchy integral theorem 3.55.

(b) A single intersection: (3.51) follows from Corollary 3.66 and Theorem 4.9.

(c) An edge, say $\overline{z_1 z_2}$, as a subset of $\mathbb{R}$: (3.51) follows from setting $\overline{w_1 w_2} := \overline{z_1 z_2} + \mathbf{i}\epsilon$ for $\epsilon \in \mathbb{R}^+$ and arguing that

$$\begin{aligned}
&\lim_{\overline{w_1 w_2} \to \overline{z_1 z_2}} \left(\int_{\overline{w_1 w_2}} f(z)\mathrm{d}z - \int_{\overline{z_1 z_2}} f(z)\mathrm{d}z \right) \\
&= \lim_{\epsilon \to 0} \int_{\overline{z_1 z_2}} (f(z + \mathbf{i}\epsilon) - f(z))\mathrm{d}z \\
&= \int_{\overline{z_1 z_2}} \lim_{\epsilon \to 0} (f(z + \mathbf{i}\epsilon) - f(z))\mathrm{d}z = 0,
\end{aligned}$$

where the second step follows from Lemma 3.101 and the last from Heine's theorem 3.57 and Lemma 3.7.

(d) All other cases: we decompose $\Delta(z_1, z_2, z_3)$ into a finite number of triangles and apply (b) and (c). □

Theorem 4.79 (Schwarz reflection principle 1867)**.** Let $\mathcal{D}$ be a domain symmetric with respect to the real axis. Define three subsets of $\mathcal{D}$ as

$$\mathcal{D}_+ := \{z \in \mathcal{D} : \operatorname{Im} z > 0\}, \tag{4.22a}$$
$$\mathcal{D}_- := \{z \in \mathcal{D} : \operatorname{Im} z < 0\}, \tag{4.22b}$$
$$\mathcal{D}_0 := \mathcal{D} \cap \mathbb{R}. \tag{4.22c}$$

If f is continuous on $\mathcal{D}_+ \cup \mathcal{D}_0$, f is analytic on $\mathcal{D}_+$, and $f(\mathcal{D}_0) \subset \mathbb{R}$, then f can be extended to an analytic function $F : \mathcal{D} \to \mathbb{C}$ given by

$$F(z) := \begin{cases} f(z) & \text{if } z \in \mathcal{D}_+ \cup \mathcal{D}_0, \\ \overline{f(\overline{z})} & \text{if } z \in \mathcal{D}_-. \end{cases} \tag{4.23}$$

Proof. By $f(z)$ being analytic on $\mathcal{D}_+$, we have, $\forall z \in \mathcal{D}_-$,

$$\begin{aligned}
&\lim_{h \to 0} \frac{F(z+h) - F(z)}{z + h - z} = \lim_{h \to 0} \frac{\overline{f\left(\overline{z} + \overline{h}\right)} - \overline{f(\overline{z})}}{h} \\
&= \lim_{h \to 0} \frac{\overline{f\left(\overline{z} + \overline{h}\right) - f(\overline{z})}}{h} = \lim_{h \to 0} \overline{\frac{f(\overline{z} + \overline{h}) - f(\overline{z})}{\overline{h}}} = \overline{f'(\overline{z})},
\end{aligned}$$

where the last step follows from $\overline{h} \to 0 \Leftrightarrow h \to 0$ for $h \in \mathbb{C}$ and the third equality follows from $\overline{\left(\frac{u}{v}\right)} = \frac{\overline{u}}{\overline{v}}$, which can be easily shown by Corollary 1.62. Then the proof is completed by Lemma 4.78. □

Corollary 4.80. If $\mathcal{D} \subseteq \mathbb{C}$ is a domain with $\mathcal{D} \cap \mathbb{R} \neq \emptyset$ and if $f, g : \mathcal{D} \to \mathbb{C}$ are analytic functions with $f|_{\mathcal{D}\cap\mathbb{R}} = g|_{\mathcal{D}\cap\mathbb{R}}$, then we have $f(z) = g(z)$ for all $z \in \mathcal{D}$.

Proof. This follows directly from the identity theorem 4.35. □

Corollary 4.81. Let $r > 0$ and $\mathcal{D}$ be a domain symmetric with respect to the circle $\partial\mathcal{U}_r(0) = \{z \in \mathbb{C} : |z| = r\}$, that is, $z \in \mathcal{D}$ if and only if $\frac{r^2}{\overline{z}} \in \mathcal{D}$. Define three subsets of $\mathcal{D}$ as

$$\mathcal{D}_+ := \mathcal{D} \cap \mathcal{U}_r(0), \tag{4.24a}$$
$$\mathcal{D}_- := \{z \in \mathcal{D} : |z| > r\}, \tag{4.24b}$$
$$\mathcal{D}_0 := \mathcal{D} \cap \partial\mathcal{U}_r(0). \tag{4.24c}$$

If f is continuous on $\mathcal{D}_+ \cup \mathcal{D}_0$, f is analytic on $\mathcal{D}_+$, and there exists $\rho > 0$ such that for all $z \in \mathcal{D}_0$, $|f(z)| = \rho$, then f can be extended to an analytic function $F : \mathcal{D} \to \mathbb{C}$ given by

$$F(z) := \begin{cases} f(z) & \text{if } z \in \mathcal{D}_+ \cup \mathcal{D}_0, \\ \rho^2 / \overline{f\left(\frac{r^2}{\overline{z}}\right)} & \text{if } z \in \mathcal{D}_-. \end{cases} \tag{4.25}$$

Exercise 4.82. Prove Corollary 4.81 by similar arguments as in the proof of Theorem 4.79.

Exercise 4.83. Show that an analytic function $f : \mathbb{E} \to \mathbb{C}$ that is continuous on $\overline{\mathbb{E}}$ must be a constant if

$$\begin{cases} f(z) \neq 0 & \text{if } z \in \mathbb{E}, \\ |f(z)| = 1 & \text{if } z \in \partial\mathbb{E}. \end{cases}$$

Definition 4.84. A *power series in two variables* centered at a point (x_0, y_0) is a power series of the form

$$f(x, y) = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} c_{n,m} (x - x_0)^n (y - y_0)^m. \tag{4.26}$$

Definition 4.85. A function $f : \mathcal{D} \to \mathbb{R}$ with $\mathcal{D} \subseteq \mathbb{R}^2$ open is *real analytic* iff for each $(x_0, y_0) \in \mathcal{D}$ there exists $r > 0$ such that for any $(x, y) \in \mathcal{U}_r(x_0, y_0)$ there exists a power series in two variables that equals $f(x, y)$, i.e., (4.26) holds.

Exercise 4.86. Show that the real part and the imaginary part of an analytic function is real analytic in the sense of Definition 4.85.

Lemma 4.87. Let $\mathcal{M} \subset \mathbb{R}$ be a non-degenerate interval, i.e., an interval that contains more than one point. A function $f : \mathcal{M} \to \mathbb{R}$ has an analytic continuation $F : \mathcal{D} \to \mathbb{C}$ on a domain $\mathcal{D} \subset \mathbb{C}$ if and only if f is real analytic in the sense of Definition B.102.

Proof. For necessity, F being analytic, Theorem 4.1, and $\mathcal{M} \subset \mathcal{D}$ yield

$$\forall c \in \mathcal{M}, \exists r > 0 \text{ s.t. } \forall z \in \mathcal{U}_r(c) \subset \mathcal{D},\ F(z) = \sum_{n=0}^{\infty} a_n (z-c)^n.$$

Then Definition 4.76 gives $F|_{\mathcal{M}} = f$, i.e.,

$$\forall x \in \mathcal{M} \cap \mathcal{U}_r(c),\ f(x) = \sum\nolimits_{n=0}^{\infty} a_n (x-c)^n.$$

Hence f is real analytic in the sense of Definition B.102.

As for the sufficiency, f being real analytic gives

$$\forall c \in \mathcal{M}, \exists r_c > 0 \text{ s.t. } \forall x \in \mathcal{M} \cap \mathcal{U}_{r_c}(c),\ f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n.$$

Define a series $F_c(z) := \sum_{n=0}^{\infty} a_n (z-c)^n$. Abel's Theorem 1.77 dictates that $F_c(z)$ converges to some complex number for any given $z \in \mathcal{U}_{r_c}(c)$. By Theorem 4.9 (A) $\Leftrightarrow$ (F), $F_c(z) : \mathcal{U}_{r_c}(c) \to \mathbb{C}$ is analytic. For each $c \in \mathcal{M}$, the above construction gives an analytic function F_c.

Starting from $F|_{c_0} = F_{c_0}$ for any $c_0 \in \mathcal{M}$, we choose c_1 sufficiently close to c_0 so that $I_{0,1} := \mathcal{U}_{r_{c_1}}(c_1) \cap \mathcal{U}_{r_{c_0}}(c_0) \neq \emptyset$. The two analytic functions F_{c_0} and F_{c_1} agree with each other on the infinite set $\mathcal{M} \cap I_{0,1}$. By the identity theorem 4.35 and Definition 4.76, the function $F_{\{0,1\}} : \mathcal{U}_{r_{c_1}}(c_1) \cup \mathcal{U}_{r_{c_0}}(c_0) \to \mathbb{C}$ is an analytic continuation of F_{c_0}. Then the proof is completed by repeating the above analytic continuation so that $\mathcal{M}$ is covered by $\mathcal{D} := \cup_{i \in \mathbb{N}} \mathcal{U}_{r(c_i)}(c_i)$ and F is given by $F := F_{\mathbb{N}}$. □

Corollary 4.88. Let $\mathcal{D} \subset \mathbb{C}$ be a domain, $\mathcal{M} \subset \mathcal{D}$ a subset with at least one limit point in $\mathcal{D}$, and $f : \mathcal{M} \to \mathbb{C}$ an analytic function. If there exists a function $F : \mathcal{D} \to \mathbb{C}$ as the analytic continuation of f, then F is unique.

Proof. This follows from the identity theorem 4.35. □

4.5 Isolated singularities of analytic functions

Example 4.89. Functions like $\frac{\sin z}{z}$, $\frac{1}{z}$, and $\exp(\frac{1}{z})$ are not defined at $z = 0$, but are all analytic in some punctured disk $\dot{\mathcal{U}}_r(0)$, cf. (1.22).

Definition 4.90. An *(isolated) singularity* of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open is a point $c \notin \mathcal{D}$ such that $\mathcal{D}$ contains a punctured disk at c, i.e.,

$$\exists r > 0 \text{ s.t. } \dot{\mathcal{U}}_r(c) \subseteq \mathcal{D}.$$

Example 4.91. The singularity $z = 0$ of the analytic function $f(z) = \frac{1}{\sin \frac{1}{z}}$ is not isolated because it is the limit point of the sequence $(\frac{1}{n\pi})_{n \in \mathbb{N}^+}$, of isolated singularities of f.

4.5.1 Classifying singularities

Definition 4.92. A singularity $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open is *removable* iff f can be analytically extended to $\mathcal{D} \cup \{c\}$, i.e. iff there exists an analytic continuation $F : \mathcal{D} \cup \{c\} \to \mathbb{C}$ with $F|_{\mathcal{D}} = f$.

Theorem 4.93 (Riemann removability condition 1851)**.** A singularity $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open is removable if and only if there exists $r > 0$ such that $\dot{\mathcal{U}}_r(c) \subseteq \mathcal{D}$ and f is bounded over $\dot{\mathcal{U}}_r(c)$.

Proof. The necessity is obvious and we only prove the sufficiency. WLOG, we assume $c = 0$. Define a function $h : \mathcal{U}_r(0) \to \mathbb{C}$ by

$$h(z) = \begin{cases} z^2 f(z), & z \neq 0; \\ 0, & z = 0. \end{cases}$$

h is analytic at 0 because $f(z)$ is bounded at $z = 0$ and

$$\lim_{z \to 0} \frac{h(z) - h(0)}{z - 0} = \lim_{z \to 0} z f(z) = 0.$$

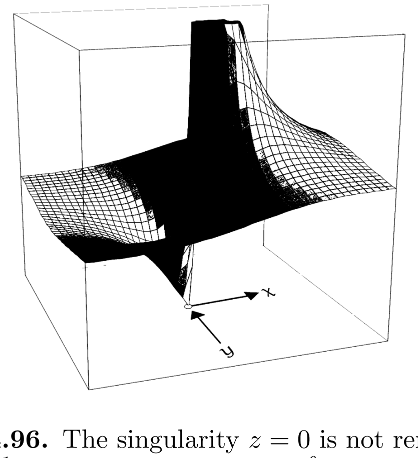
Thus h is analytic in $\mathcal{U}_r(0)$ and Theorem 4.1 yields

$$h(z) = a_0 + a_1 z + a_2 z^2 + \cdots$$

Then $h(0) = h'(0) = 0$ implies $a_0 = a_1 = 0$. For all $z \neq 0$, the power series $a_2 + a_3 z + a_4 z^2 + \cdots$ defines an analytic function F that satisfies $F|_{\dot{\mathcal{U}}_r(0)} = f$ and is also analytic at $z = 0$. The proof is then completed by Definition 4.92. □

Example 4.94. Since $\lim_{z \to 0} \frac{\sin z}{z} = 1$, Theorem 4.93 implies that the singularity $z = 0$ of $f(z) = \frac{\sin z}{z}$ is removable.

Example 4.95. The singularity $z = 0$ is not removable for $f(z) = \frac{1}{z}$.



Example 4.96. The singularity $z = 0$ is not removable for $g(z) = \exp(\frac{1}{z})$. Since $|g(z)| = \exp(\frac{\cos\theta}{r})$, the behavior of g at 0 depends upon the direction in which we approach 0: $\cos\theta < 0$ implies $|g(z)| \to 0$; $\cos\theta = 0$ implies $|g(z)| \to 1$; and $\cos\theta > 0$ implies $|g(z)| \to +\infty$ as a rate faster than any polynomial! See the plot above.

Exercise 4.97. Let $c \notin \mathcal{D}$ be a singularity of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open. Show that the following statements are equivalent:

(a) c is a removable singularity of f.

(b) There exists $r > 0$ such that $\dot{\mathcal{U}}_r(c) \subseteq \mathcal{D}$ and f is bounded over $\dot{\mathcal{U}}_r(c)$.

(c) The limit $\lim_{z\to c} f(z)$ exists.

(d) $\lim_{z\to c}(z-c)f(z) = 0$.

Exercise 4.98. Is 0 a removable singularity of $g(z) = \sin\frac{1}{z}$? Why?

Exercise 4.99. Let $c \notin \mathcal{D}$ be a singularity of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open. Show that c is removable singularity of f if there exist positive constants r, A, and ϵ such that

$$\forall z \in \dot{\mathcal{U}}_r(c), \qquad |f(z)| \le A|z-c|^{-1+\epsilon}.$$

Definition 4.100. A singularity $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open is said to be *non-essential* iff there exists an integer $m \in \mathbb{Z}$ such that c is a removable singularity of

$$g(z) := (z-c)^m f(z). \tag{4.27}$$

In particular, a *pole* is a non-essential singularity that is not removable.

Definition 4.101. A singularity of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open is said to be *essential* if it is not non-essential.

Lemma 4.102. Let $c \notin \mathcal{D}$ be a non-essential singularity of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open. If f does not vanish identically in some neighborhood of c, then there exists a minimal integer $k \in \mathbb{Z}$ such that the function

$$h(z) = (z-c)^k f(z) \tag{4.28}$$

has a removable singularity at c and $h(c) \neq 0$.

Proof. If c is a non-essential singularity of f, then by Definitions 4.100 and 4.92 the function $g(z)$ in (4.27) can be expanded as a power series at c. Furthermore, f does not vanish identically in some neighborhood of c, hence

$$\begin{aligned}&\exists n \in \mathbb{N} \text{ s.t. } a_n \neq 0 \text{ and}\\ &\quad g(z) = a_n(z-c)^n + a_{n+1}(z-c)^{n+1} + \cdots \end{aligned}\tag{4.29}$$

Multiply both sides with $(z-c)^{-n}$, refer to (4.27), and we see that the function $h(z)$ in (4.28) has a removable singularity at c with $k = m-n$. In addition, $h(c) \neq 0$ follows from $a_n \neq 0$. Furthermore, k is the minimal integer with this property. Otherwise the power series

$$(z-c)^{k-1}f(z) = \frac{a_n}{z-c} + a_{n+1} + a_{n+2}(z-c) + \cdots$$

would have a removable singularity. But this is false because Theorem 4.93 dictates that the singularity c of $\frac{a_n}{z-c}$ is not removable. $\square$

Definition 4.103. The *order of a non-essential singularity* c of an analytic function f, written $\operatorname{ord}(f;c)$, is the maximum number $m \in \mathbb{Z}$ such that $h(z) := (z-c)^{-m}f(z)$ has a removable singularity at c and $h(c) \neq 0$. In particular,

$$\operatorname{ord}(f;c) = +\infty \iff \exists r > 0 \text{ s.t. } \forall z \in \mathcal{U}_r(c), f(z) = 0.$$

Example 4.104. The analytic function $f : \mathbb{C} \setminus \{1\} \to \mathbb{C}$,

$$f(z) = (z-1)^5 + 2(z-1)^6 = (z-1)^5(1+2(z-1)),$$

has a non-essential singularity at $z = 1$ of order 5. By Definition 4.18 and Lemma 4.20, $z = 1$ is a zero point of f with multiplicity 5. This suggests that the order of a non-essential singularity reduces to the multiplicity of a zero point for $\operatorname{ord}(f;c) > 0$.

Lemma 4.105. A non-essential singularity c of an analytic function f is removable if and only if $\operatorname{ord}(f;c) \ge 0$.

Proof. We show that $m := \operatorname{ord}(f;c) \ge 0$ implies removability and $m < 0$ implies non-removability. Since c is a non-essential singularity of order m, Definition 4.103 and Theorem 4.93 yield

$$\exists C_m > 0, \exists r > 0, \text{ s.t. } \forall z \in \dot{\mathcal{U}}_r(c),\ |f(z)(z-c)^{-m}| \in (0, C_m).$$

When $m = 0$, the above equation reduces to the condition of f being bounded in $\dot{\mathcal{U}}_r(c)$, and then Theorem 4.93 implies that c is removable. For any given $m > 0$, c is also removable by Theorem 4.93 because

$$\begin{aligned}&\exists C = C_m, \exists r = \min(1, r_m), \text{ s.t. } \forall z \in \dot{\mathcal{U}}_r(c),\\ &\quad |f(z)| = \left|f(z)(z-c)^{-m}\right| \cdot |z-c|^m < C_m = C,\end{aligned}$$

where the inequality follows from $|z-c| < 1$ and $m > 0$.

For any given $m < 0$, let $h(z) = (z-c)^{-m}f(z)$. Then Definition 4.103 implies that $h(c) \neq 0$. The continuity of h at c then yields

$$\exists \rho > 0 \text{ s.t. } \forall z \in \dot{\mathcal{U}}_\rho(c),\ 0 \neq |h(z)| \ge h_m := \inf_{z \in \dot{\mathcal{U}}_\rho(c)} |h(z)|.$$

It follows that $f(z)$ is not bounded at c because

$$\begin{aligned}&\forall C > 0, \exists r = \min\left(\rho, \frac{h_m^{-\frac{1}{m}}}{C^{-\frac{1}{m}}}\right) \text{ s.t. } \forall z \in \mathcal{U}_r(c),\\ &\quad |f(z)| = \frac{|h(z)|}{|z-c|^{-m}} \ge \frac{h_m}{|z-c|^{-m}} > \frac{h_m}{r^{-m}} \ge C.\end{aligned} \tag{4.30}$$

Then Theorem 4.93 completes the proof. $\square$

Definition 4.106. The *order of a pole* c is the negative of its order as a non-essential singularity. In particular, a pole of order 1 is called a *simple pole*.

Corollary 4.107. A non-essential singularity c of an analytic function f is a pole if and only if its order is negative.

Proof. This follows from Definitions 4.100 and 4.103 and Lemma 4.105. $\square$

Exercise 4.108. Let $c \notin \mathcal{D}$ be a singularity of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open. Show that the following statements are equivalent:

(a) c is a pole of f of order $k \in \mathbb{N}$.

(b) There exist $r > 0$ and an analytic function $h : \mathcal{U}_r(c) \to \mathbb{C}$ with $h(c) \neq 0$ such that $\dot{\mathcal{U}}_r(c) \subseteq \mathcal{D}$ and

$$\forall z \in \dot{\mathcal{U}}_r(c), \qquad f(z) = \frac{h(z)}{(z-c)^k}.$$

(c) There exist $r > 0$ and an analytic function $g : \mathcal{U}_r(c) \to \mathbb{C}$ not vanishing in $\dot{\mathcal{U}}_r(c)$ such that g has a zero of multiplicity k at c and $f = \frac{1}{g}$ over $\dot{\mathcal{U}}_r(c)$.

(d) There exist $r > 0$ and constants $M_1, M_2 > 0$ such that

$$\forall z \in \dot{\mathcal{U}}_r(c), \qquad M_1|z-c|^{-k} \le |f(z)| \le M_2|z-c|^{-k}.$$

Example 4.109. The analytic function $f : \mathbb{C}^\bullet \to \mathbb{C}$,

$$f(z) = \frac{1}{z^2} + \frac{1}{z} = \frac{1+z}{z^2}$$

has a pole at $z = 0$ of order 2.

Corollary 4.110. At a pole $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open, we have

$$\lim_{z\to c; z\in\mathcal{D}} |f(z)| = \infty, \tag{4.31}$$

i.e., $\forall C > 0, \exists r > 0$ s.t. $\forall z \in \mathcal{D}$, $|z-a| < r \implies |f(z)| \ge C$.

Proof. By Corollary 4.107, a pole is a non-essential singularity with a negative order. The proof is then completed by repeating the arguments in proving (4.30). □

Lemma 4.111. Suppose $c \notin \mathcal{D}$ is a non-essential singularity of analytic functions $f, g : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open. Then c is also a non-essential singularity of the functions

$$f \pm g,\ f\cdot g,\ \frac{f}{g} \text{ satisfying } \forall z \in \mathcal{D},\ g(z) \neq 0,$$

and we have

$$\operatorname{ord}(f \pm g; c) \ge \min\{\operatorname{ord}(f;c), \operatorname{ord}(g;c)\}, \tag{4.32a}$$

$$\operatorname{ord}(f\cdot g; c) = \operatorname{ord}(f;c) + \operatorname{ord}(g;c), \tag{4.32b}$$

$$\operatorname{ord}\left(\frac{f}{g}; c\right) = \operatorname{ord}(f;c) - \operatorname{ord}(g;c). \tag{4.32c}$$

Proof. By Definition 4.103, we write $f(z) = (z-c)^m h(z)$ with $h(c) \neq 0$ and $g(z) = (z-c)^n \ell(z)$ with $\ell(c) \neq 0$. The rest of the proof follows trivially. □

Theorem 4.112 (Casorati-Weierstrass)**.** For any neighborhood $\dot{\mathcal{U}}_r(c)$ of an essential singularity $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open, the image $f(\dot{\mathcal{U}}_r(c)\cap\mathcal{D})$ is dense in $\mathbb{C}$ in the sense of Definition D.19.

Proof. By Definition D.19, A being dense in $\mathbb{C}$ means

$$\forall x \in \mathbb{C}, \forall \epsilon > 0,\ A \cap \mathcal{U}_\epsilon(x) \neq \emptyset. \tag{4.33}$$

Suppose there exists $r > 0$ such that $f(\dot{\mathcal{U}}_r(c) \cap \mathcal{D})$ is not dense in $\mathbb{C}$. Negating (4.33) with $A = f(\dot{\mathcal{U}}_r(c) \cap \mathcal{D})$ gives

$$\exists b \in \mathbb{C}\ \exists \epsilon > 0 \text{ s.t. } f(\dot{\mathcal{U}}_r(c) \cap \mathcal{D}) \cap \mathcal{U}_\epsilon(b) = \emptyset.$$

The analytic function $g(z) := \frac{1}{f(z)-b}$ is bounded in $\dot{\mathcal{U}}_\delta(c)$ for some $\delta \in (0, r)$. Theorem 4.93 then implies that $z = c$ is a removable singularity of $g(z)$, which contradicts the fact that c is an essential singularity of f:

- if $g(c) \neq 0$, $f(c) = \frac{1}{g(c)} + b$ is bounded and thus c is a removable singularity of f;
- if $g(c) = 0$, c is a root of g with multiplicity m. Thus $\lim_{z\to c} f(z) = \lim_{z\to c} \frac{1}{g(z)} + b = \infty$. By Definitions 4.100 and 4.106, c is a pole of f of order m. □

Theorem 4.113 (Classifying isolated singularities by their mapping behaviors)**.** A singularity $c \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open must be one of the three types as follows.

(a) c is removable $\Leftrightarrow$ f is bounded in $\dot{\mathcal{U}}_r(c)$ for some $r > 0$;

(b) c is a pole $\Leftrightarrow$ $\lim_{z\to c; z\in\mathcal{D}} |f(z)| = \infty$;

(c) c is essential $\Leftrightarrow$ $\forall r > 0$, $f(\dot{\mathcal{U}}_r(c) \cap \mathcal{D})$ is dense in $\mathbb{C}$.

Proof. By Definitions 4.100 and 4.101, a singularity must be one of the three types. We need to prove the three equivalences. (a) is just Theorem 4.93. If $\lim_{z\to c; z\in\mathcal{D}} |f(z)| = \infty$, then c is neither removable nor essential, hence it must be a pole. Then (b) holds from Corollary 4.110. If $f(\dot{\mathcal{U}}_r(c) \cap \mathcal{D})$ is dense in $\mathbb{C}$, then by Theorem 4.93 c is not removable. Also, $\lim_{z\to c; z\in\mathcal{D}} |f(z)| = \infty$ does not hold; otherwise for any $c \in \mathbb{C}$ there exists $r_c > 0$ small enough so that the distance from c to $f(\dot{\mathcal{U}}_{r_c}(c) \cap \mathcal{D})$ is arbitrarily large. The proof is completed by Theorem 4.112. □

Exercise 4.114. Find all isolated singularities of

$$f(z) = \frac{(z-1)^2(z+3)}{1 - \sin\frac{\pi z}{2}}$$

and determine for each one its type.

4.5.2 Laurent decomposition

Notation 6. For $r \in [0, +\infty)$ and $R \in (r, +\infty]$, the *annulus centered at c with radii r, R* is

$$\mathcal{A}_c(r, R) := \{z \in \mathbb{C} : r < |z-c| < R\}. \tag{4.34}$$

In particular, $\mathcal{A}_c(r, +\infty) = \{z \in \mathbb{C} : |z-c| > r\}$.

Theorem 4.115 (Cauchy integral formula for annuli)**.** A continuous function $f : \overline{\mathcal{A}_c(r, R)} \to \mathbb{C}$ that is also analytic on $\mathcal{A}_c(r, R)$ satisfies, $\forall z \in \mathcal{A}_c(r, R)$,

$$f(z) = \frac{1}{2\pi\mathbf{i}} \oint_{|\zeta-c|=R} \frac{f(\zeta)}{\zeta - z}\mathrm{d}\zeta - \frac{1}{2\pi\mathbf{i}} \oint_{|\zeta-c|=r} \frac{f(\zeta)}{\zeta - z}\mathrm{d}\zeta. \tag{4.35}$$

Proof. The conclusion follows as a special case of Corollary 3.76 with $n = 1$. □

Exercise 4.116. Give an alternative proof of Theorem 4.115 by applying the Riemann removability theorem 4.93 to the function $g_z : \mathcal{A}_c(r, R) \setminus \{z\} \to \mathbb{C}$ given by

$$g_z(\zeta) := \frac{f(\zeta) - f(z)}{\zeta - z}.$$

You are not supposed to use Corollary 3.76.

Theorem 4.117 (Laurent decomposition)**.** A continuous function $f : \overline{\mathcal{A}_c(r, R)} \to \mathbb{C}$ that is analytic on $\mathcal{A}_c(r, R)$ admits a decomposition as

$$f(z) = g(z) + h(z), \tag{4.36}$$

where $g : \mathcal{U}_R(c) \to \mathbb{C}$ and $h : \mathcal{A}_c(r, +\infty) \to \mathbb{C}$ are analytic. Moreover, the decomposition (4.36) is unique if $h(\infty) = 0$.

Proof. The decomposition follows from Theorem 4.115 and

$$\begin{aligned} g(z) &:= +\tfrac{1}{2\pi\mathbf{i}}\oint_{|\zeta-c|=R}\tfrac{f(\zeta)}{\zeta-z}\mathrm{d}\zeta,\\ h(z) &:= -\tfrac{1}{2\pi\mathbf{i}}\oint_{|\zeta-c|=r}\tfrac{f(\zeta)}{\zeta-z}\mathrm{d}\zeta, \end{aligned}$$

where $h(\infty)=0$. Both g and h are clearly analytic.

Suppose $f(z)=\tilde{h}(z)+\tilde{g}(z)$ is a different decomposition with $\tilde{h}(\infty)=0$. Then we have $\tilde{h}(z)+\tilde{g}(z)=h(z)+g(z)$ and the functions $H:=h-\tilde{h}$ and $G:=\tilde{g}-g$ satisfies $G(z)=H(z)$ for all $z\in\mathcal{A}_c(r,R)$. Since H is analytic on $\mathcal{A}_c(r,+\infty)$, G is analytic on $\mathcal{U}_R(c)$, $r<R$, and $\mathcal{A}_c(r,R)=\mathcal{U}_R(c)\cap\mathcal{A}_c(r,+\infty)$, the identity theorem 4.35 and Definition 4.76 imply that G and H glue to a single entire function F satisfying $\lim_{z\to\infty}F(z)=H(\infty)=h(\infty)-\tilde{h}(\infty)=0$. Liouville's theorem 3.95 dictates that both $G(z)$ and $H(z)$ be the constant zero, which gives $g=\tilde{g}$ and $h=\tilde{h}$. □

Definition 4.118. In the Laurent decomposition (4.36), the analytic functions $g:\mathcal{U}_R(c)\to\mathbb{C}$ and $h:\mathcal{A}_c(r,+\infty)\to\mathbb{C}$ are respectively called the *analytic/regular part* and the *principal part.*

4.5.3 Laurent series

Definition 4.119. A *doubly infinite series* is an infinite series of the form

$$\sum_{j=-\infty}^{+\infty}a_j=\sum_{j=0}^{+\infty}a_j+\sum_{j=1}^{+\infty}a_{-j},\tag{4.37}$$

it *converges* iff the series $\sum_{j=0}^{+\infty}a_j$ and $\sum_{j=1}^{+\infty}a_{-j}$ both converges independently.

Definition 4.120. A *Laurent series* on $\mathcal{U}_r(c)$ is a doubly infinite series of the form $\sum_{j=-\infty}^{+\infty}a_j(z-c)^j$.

Lemma 4.121. If a Laurent series $\sum_{j=-\infty}^{+\infty}a_j(z-c)^j$ converges at some $z=z_0$, then there exist unique $r,R\in[0,\infty]$ such that the series converges normally in its *annulus of convergence* $\mathcal{A}_c(r,R)$ and diverges for all $z\in\mathcal{A}_c(R,+\infty)$ and all $z\in\mathcal{A}_c(0,r)$. In addition, for r_1 and r_2 satisfying $r<r_1<r_2<R$, the Laurent series converges uniformly and absolutely on $\overline{\mathcal{A}_c(r_1,r_2)}$.

Proof. WLOG, we set $c=0$ and assume the point of convergence $z_0\neq 0$. By Definition 4.119, the convergence of $\sum_{j=-\infty}^{+\infty}a_j(z-c)^j$ at z_0 implies those of

$$p_+(z)=\sum_{j=0}^{+\infty}a_jz^j,\quad p_-\left(\frac{1}{z}\right)=\sum_{j=1}^{+\infty}a_{-j}\left(\frac{1}{z}\right)^j.\tag{4.38}$$

By the Abel theorem 1.77, there exists $r_2>|z_0|$ such that $p_+(z)$ converges normally for all $z\in\mathcal{U}_{r_2}(c)$ and there exists $r>\left|\frac{1}{z_0}\right|$ such that $p_-\left(\frac{1}{z}\right)$ converges normally for all z satisfying $\left|\frac{1}{z}\right|<r$. Set $r_1=\frac{1}{r}$ and we have $r_1<r_2$ and the first conclusion, which, together with Theorem 1.75 and $\overline{\mathcal{A}_c(r_1,r_2)}$ being compact implies the second conclusion. □

Example 4.122. Apply Lemma 4.121 to Laurent series of

$$f(z):=\frac{2}{z^2-4z+3}=\frac{1}{1-z}+\frac{1}{z-3}$$

on three annuli $\mathcal{A}_0(0,1)$, $\mathcal{A}_0(1,3)$, and $\mathcal{A}_0(3,+\infty)$.

$z\in\mathcal{A}_0(0,1)$ implies $r=0$ and $R=1$ because

$$\begin{aligned}&\tfrac{1}{1-z}=\textstyle\sum_{n=0}^{\infty}z^n \text{ and } \tfrac{1}{3-z}=\tfrac{1}{3}\cdot\tfrac{1}{1-z/3}=\sum_{n=0}^{\infty}\tfrac{z^n}{3^{n+1}}\\ &\implies g(z)=f(z)=\textstyle\sum_{n=0}^{\infty}\left(1-\tfrac{1}{3^{n+1}}\right)z^n \text{ and } h(z)=0.\end{aligned}$$

$z\in\mathcal{A}_0(1,3)$ implies $r=1$ and $R=3$ because

$$\begin{aligned}&\begin{cases}-h(z)=\frac{1}{z-1}=\frac{1}{z}\cdot\frac{1}{1-1/z}=\sum_{n=0}^{\infty}\frac{1}{z^{n+1}},\\ -g(z)=\frac{1}{3-z}=\sum_{n=0}^{\infty}\frac{z^n}{3^{n+1}}\end{cases}\\ &\implies f(z)=g(z)+h(z)=-\textstyle\sum_{n=1}^{\infty}\frac{1}{z^n}-\sum_{n=0}^{\infty}\frac{z^n}{3^{n+1}}.\end{aligned}$$

$z\in\mathcal{A}_0(3,+\infty)$ implies $r=3$ and $R=+\infty$ because

$$\begin{aligned}&\tfrac{1}{z-1}=\textstyle\sum_{n=1}^{\infty}\frac{1}{z^n} \text{ and } \tfrac{1}{z-3}=\tfrac{1}{z}\cdot\tfrac{1}{1-3/z}=\sum_{n=0}^{\infty}\tfrac{3^n}{z^{n+1}}\\ &\implies h(z)=f(z)=\textstyle\sum_{n=1}^{\infty}\left(3^{n-1}-1\right)\frac{1}{z^n} \text{ and } g(z)=0.\end{aligned}$$

Exercise 4.123. Find all possible Laurent series of

$$f(z):=\frac{1}{z(z-1)(z-2)}.$$

Theorem 4.124 (Laurent series)**.** An analytic function $f:\mathcal{A}_c(r,R)\to\mathbb{C}$ can be represented by a Laurent series:

$$\forall z\in\mathcal{A}_c(r,R),\quad f(z)=\sum_{j=-\infty}^{+\infty}a_j(z-c)^j,\tag{4.39}$$

which converges absolutely and uniformly on each compact subset of $\mathcal{A}_c(r,R)$ with each coefficient uniquely given by

$$\forall j\in\mathbb{Z},\ a_j=\frac{1}{2\pi\mathbf{i}}\oint_{|\zeta-c|=\rho}\frac{f(\zeta)}{(\zeta-c)^{j+1}}\mathrm{d}\zeta,\tag{4.40}$$

which does not depend on $\rho\in(r,R)$ and satisfies

$$|a_j|\le\frac{1}{\rho^j}\sup\{f(\zeta):|\zeta-c|=\rho\}.\tag{4.41}$$

Proof. For any $\forall z\in\mathcal{A}_c(r,R)$, there exists r_1 and r_2 satisfying $r<r_1<r_2<R$ such that $z\in\mathcal{A}_c(r_1,r_2)$. By Theorems 4.115 and 4.117, the following decomposition is unique:

$$f(z)=\frac{1}{2\pi\mathbf{i}}\oint_{|\zeta-c|=r_2}\frac{f(\zeta)}{\zeta-z}\mathrm{d}\zeta-\frac{1}{2\pi\mathbf{i}}\oint_{|\zeta-c|=r_1}\frac{f(\zeta)}{\zeta-z}\mathrm{d}\zeta.\tag{4.42}$$

Repeat the proof of Taylor expansions in Theorem 4.1 for the first RHS integral in (4.42) and we have

$$\oint_{|\zeta-c|=r_2}\frac{f(\zeta)}{\zeta-z}\mathrm{d}\zeta=\sum_{j=0}^{+\infty}\left(\oint_{|\zeta-c|=r_2}\frac{f(\zeta)}{(\zeta-c)^{j+1}}\mathrm{d}\zeta\right)(z-c)^j.$$

For the second RHS integral in (4.42), we have

$$\begin{aligned}\oint_{|\zeta-c|=r_1}\frac{f(\zeta)}{\zeta-z}\mathrm{d}\zeta&=-\oint_{|\zeta-c|=r_1}\frac{f(\zeta)}{1-\frac{\zeta-c}{z-c}}\cdot\frac{1}{z-c}\mathrm{d}\zeta\\ &=-\oint_{|\zeta-c|=r_1}\frac{f(\zeta)}{z-c}\sum_{j=0}^{+\infty}\frac{(\zeta-c)^j}{(z-c)^j}\mathrm{d}\zeta\\ &=-\sum_{j=-\infty}^{-1}\left[\oint_{|\zeta-c|=r_1}\frac{f(\zeta)}{(\zeta-c)^{j+1}}\mathrm{d}\zeta\right](z-c)^j.\end{aligned}$$

Consequently, the a_j's for non-negative and negative j's have the same form except the domain integration:

$$\forall j \in \mathbb{N},\ a_j = \frac{1}{2\pi \mathbf{i}} \oint_{|\zeta - c| = r_2} \frac{f(\zeta)}{(\zeta - c)^{j+1}} \mathrm{d}\zeta, \tag{4.43a}$$

$$\forall j \in -\mathbb{N}^+,\ a_j = \frac{1}{2\pi \mathbf{i}} \oint_{|\zeta - c| = r_1} \frac{f(\zeta)}{(\zeta - c)^{j+1}} \mathrm{d}\zeta. \tag{4.43b}$$

The Cauchy integral theorem implies that we can set r_1 and r_2 in (4.43) to any $\rho \in (r, R)$ without changing the value of a_j's. The uniqueness of a_j's in (4.40) then follows from the uniqueness of (4.43) and Theorem 4.1. Finally, (4.41) follows from a variable substitution and Lemma 3.7. □

Example 4.125. The Laurent series of the function $f(z) = \exp(-\frac{1}{z^2})$ at the essential singularity $z = 0$ is obtained from the power series of $\exp(w)$ with $w = -\frac{1}{z^2}$:

$$f(z) = 1 - \frac{1}{z^2} + \frac{1}{2!}\frac{1}{z^4} - \frac{1}{3!}\frac{1}{z^6} + \ldots = 1 + h(z),$$

where $h(z)$ is the principal part of $f(z)$.

Exercise 4.126. Let $f : \mathcal{A}_c(r, R) \to \mathbb{C}$ be an analytic function with the Laurent expansion (4.39). Prove that

$$\forall \rho \in (r, R), \quad \sum_{j=-\infty}^{+\infty} |a_j|^2 \rho^{2j} \le [M(\rho)]^2,$$

where $M(\rho)$ is defined in Corollary 3.86. In particular, for any $j \in \mathbb{Z}$, $|a_j| \le \rho^{-j} M(\rho)$; the equality holds for some $j = j_0$ and $\rho = \rho_0$ if and only if $f(z) = a_{j_0}(z - c)^{j_0}$.

Lemma 4.127. The Laurent expansion of an analytic function $f : \mathcal{A}_0(r, R) \to \mathbb{C}$ only contains even (or odd) powers of z if f is respectively even (or odd).

Exercise 4.128. Prove Lemma 4.127.

Corollary 4.129. For a singularity c of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open, there exists $R > 0$ such that f has a unique Laurent expansion at c in $\mathcal{A}_c(0, R)$, i.e.,

$$\forall z \in \mathcal{A}_c(0, R), \quad f(z) = \sum_{j=-\infty}^{+\infty} a_j (z - c)^j. \tag{4.44}$$

Proof. This follows directly from Theorem 4.124 and the fact that a punctured disk is an annulus with zero inner radius and finite outer radius. □

Theorem 4.130 (Classifying isolated singularities by Laurent series)**.** According to (4.44), a singularity c of an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open is

(a) removable iff
$$\forall j < 0, \quad a_j = 0, \tag{4.45}$$

(b) a pole of order $k \in \mathbb{N}^+$ iff
$$a_{-k} \ne 0 \ \text{ and } \ \forall n < -k,\ a_n = 0, \tag{4.46}$$

(c) essential iff neither (4.45) nor (4.46) holds.

Proof. By (4.45), the function f can be expressed as a power series, thus it is analytic at some punctured neighborhood of c and thus bounded at this punctured neighborhood. Then it follows from Theorem 4.93 that (4.45) is a necessary and sufficient condition of c being a removable singularity. The condition (4.46) is equivalent to (4.31), then it follows from Theorem 4.113 (b) that (4.46) is a necessary and sufficient condition of c being a pole. The rest of the proof follows from Definitions 4.100 and 4.101. □

Exercise 4.131. Give an alternative proof of (d) ⇒ (a) in Exercise 4.97 by applying Exercise 4.126.

Definition 4.132. A function $f : \mathcal{S}_{a,b} \to \mathbb{C}$ with

$$\mathcal{S}_{a,b} := \{z = x + y\mathbf{i} \in \mathbb{C} : y \in (a, b)\} \tag{4.47}$$

and $-\infty \le a < b \le +\infty$ is *1-periodic* iff $f(z + 1) = f(z)$ holds for any $z \in \mathcal{S}_{a,b}$.

Corollary 4.133 (Fourier series of a 1-periodic analytic function)**.** Any 1-periodic analytic function $f : \mathcal{S}_{a,b} \to \mathbb{C}$ as in Definition 4.132 can be represented as a uniquely normally convergent *complex Fourier series*

$$f(z) = \sum_{n=-\infty}^{+\infty} a_n e^{2\pi \mathbf{i} n z}, \tag{4.48}$$

where $z = x + \mathbf{i}y$ and the *Fourier coefficients* are given by

$$a_n = \int_0^1 f(z) e^{-2\pi \mathbf{i} n z} \mathrm{d}x. \tag{4.49}$$

Proof. Write $r := e^{-2\pi b}$ and $R := e^{-2\pi a}$, construct a conformal map $q : \mathcal{S}_{a,b} \to \mathcal{A}_0(r, R)$ by $q(z) := e^{2\pi \mathbf{i} z}$, define an analytic function $g : \mathcal{A}_0(r, R) \to \mathbb{C}$ by $f = g \circ q$, and we can represent f by a Laurent series

$$f(z) = g(q(z)) = \sum_{n=-\infty}^{+\infty} a_n q^n.$$

By Theorem 4.124, the coefficient a_n is given by

$$\begin{aligned} a_n &= \frac{1}{2\pi \mathbf{i}} \oint_{|\zeta| = \rho} \frac{g(\zeta)}{\zeta^{n+1}} \mathrm{d}\zeta \\ &= \int_0^{\frac{g(\rho e^{2\pi \mathbf{i} x})}{\rho^n e^{2\pi \mathbf{i} n x}}} \mathrm{d}x = \int_0^1 f(z) e^{-2\pi \mathbf{i} n z} \mathrm{d}x, \end{aligned}$$

where the last step follows from $\rho = e^{-2\pi y}$. □

Exercise 4.134 (Laurent series as an expansion in an orthonormal basis)**.** Denote by $\mathcal{H}$ the space of all analytic functions on $\mathcal{A}_0(r, R)$ where $r < 1 < R$ and define a sesquilinear map $\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{C}$,

$$\langle f, g \rangle := \frac{1}{2\pi \mathbf{i}} \oint_{|\zeta| = 1} f(\zeta) \overline{g(\zeta)} \frac{1}{\zeta} \mathrm{d}\zeta. \tag{4.50}$$

Show that

(a) $\langle \cdot, \cdot \rangle$ is an inner product on $\mathcal{H}$ in the sense of Definition C.170;

(b) the set $\{z^j : j \in \mathbb{Z}\}$ is an orthonormal basis of $\mathcal{H}$;

(c) the Laurent series of any $f \in \mathcal{H}$ is exactly the expansion of f in terms of the orthonormal basis $\{z^j : j \in \mathbb{Z}\}$. In other words, if $f(z) = \sum_{j=-\infty}^{+\infty} a_j z^j$ is the Laurent expansion of f, then

$$\forall j \in \mathbb{Z}, \quad a_j = \left\langle f, z^j \right\rangle = \frac{1}{2\pi \mathbf{i}} \oint_{|\zeta|=1} \frac{f(\zeta)}{\zeta^{j+1}} \mathrm{d}\zeta.$$